

Week 1:

1. a) Write a C program to find the sum and average of three numbers.

Algorithm:

Step 1: Start

Step 2: Declare variables num1, num2,num3 and sum,average.

Step 3: Read values num1,num2,num3

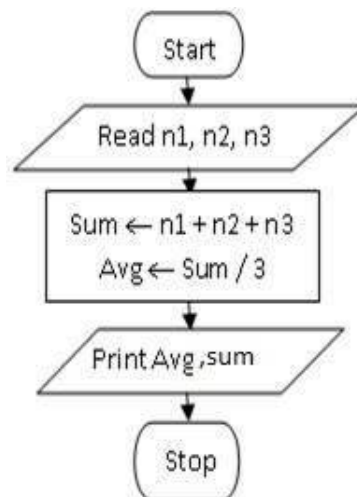
Step 4: Add num1,num2,num3 and assign the result to sum.

$\text{sum} \leftarrow \text{num1} + \text{num2} + \text{num3}$

$\text{average} \leftarrow \text{sum} / 3$

Step 5: Display sum and average

Step 6: Stop

Flow Chart:

Program:

```
#include<stdio.h>
int main( )
{
    int a,b,c;
    int sum,average;
    printf("Enter any three integers: ");
    scanf("%d%d %d",&a,&b,&c);
    sum = a+b+c;
    average=sum/3
    printf("Sum and average of three integers: %d %d",sum,average);
    return 0;
}
```

SAMPLE INPUT:

Enter any three integers:2 4 5

EXPECTED OUTPUT:

Sum and average of three integers: 11 3

Record at least 3 results

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1. b) Write a C program to find the sum of individual digits of positive integer.**AIM:**

To find the sum of individual digits of positive integer.

Description:

Summation of digits of a number

Ex: 1234

Summation = $1+2+3+4=10$

ALGORITHM:

Step 1: Start

Step 2: Read n

Step 3: Initialize sum $\leftarrow 0$

Step 4: while(n!=0)

Begin

Step 5: $r \leftarrow n \% 10$

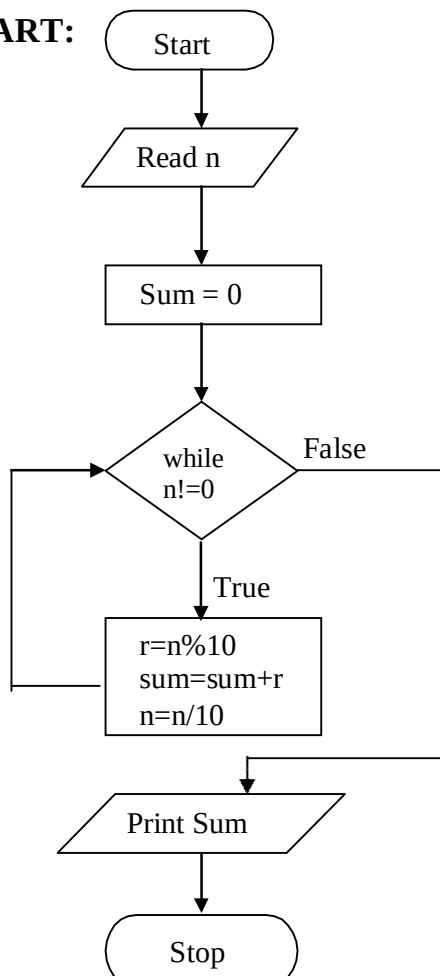
Step 6: $\text{sum} \leftarrow \text{sum} + r$

Step 7: $n \leftarrow n / 10$

End

Step 8: Print "sum"

Step 9: Stop

FLOWCHART:

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,r,sum=0;
    clrscr();
    printf("ENTER A POSITIVE INTEGER \n");
    scanf("%d",&n);
    while(n!=0)
    {
        r=n%10;
        sum=sum+r;
        n=n/10;
    }
    printf("THE SUM OF INDIVIDUAL DIGITS OF A POSITIVE INTEGER IS..%d",sum);
    getch();
}
```

SAMPLE INPUT:

ENTER A POSITIVE INTEGER
5 3 2 1

EXPECTED OUTPUT:

THE SUM OF INDIVIDUAL DIGITS OF A POSITIVE INTEGER IS..11

Record at least 3 results

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1).c) Fibonacci Sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.

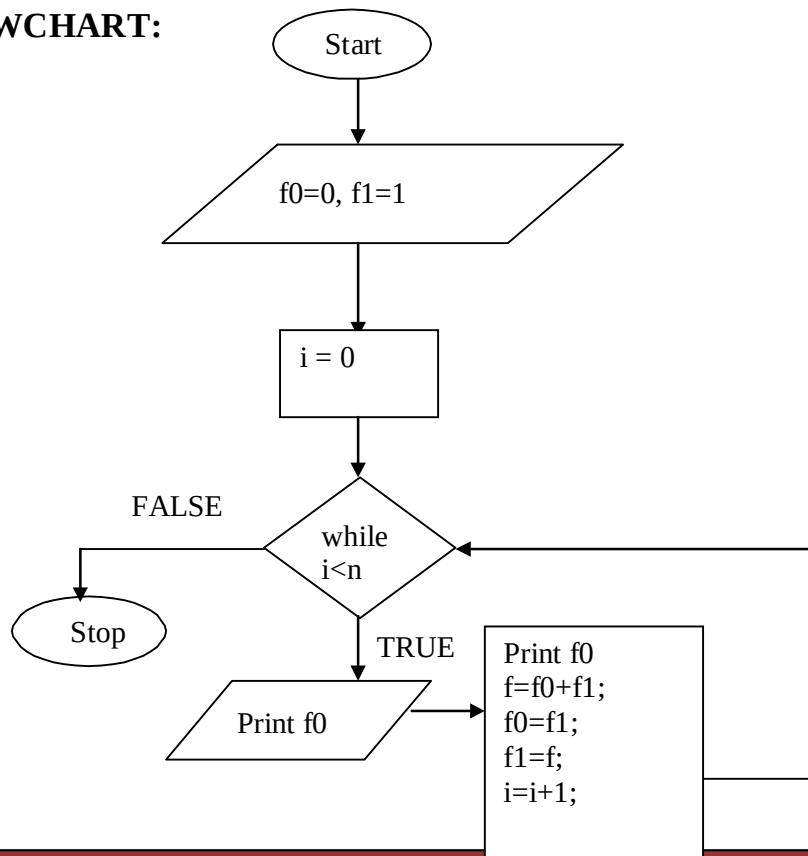
AIM: To generate the first n terms of the Fibonacci sequence..

Description: Initial Fibonacci numbers are 0 and 1. Next number can be generated by adding two numbers. So $0+1=1$. Therefore next number can be generated by adding two previous . so Fibonacci series is 0 1 1 2 3 5

ALGORITHM:

Step 1 : Start
 Step 2 : Read n
 Step 3 : Initialize $f0 \leftarrow 0$, $f1 \leftarrow 1$, $f \leftarrow 0$
 Step 4 : $i=0$
 Step 5 : while($i \leq n$) do as follows
 $\text{printf}("%d\\t", f0);$
 $f=f0+f1;$
 $f0=f1;$
 $f1=f;$
 $i=i+1;$
 If not goto step 7
 Step 6 : Stop

FLOWCHART:



PROGRAM:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int f0,f1,f,n,i;
    clrscr();
    printf("ENTER THE VALUE FOR n \n");
    scanf("%d",&n);
    f0=0;
    f1=1;
    printf("FIBONACCI SEQUENCE FOR THE FIRST %d TERMS:\n",n);
    i=0;
    while(i<n)
    {
        printf("%d\t",f0);
        f=f0+f1;
        f0=f1;
        f1=f;
        i=i+1;
    }
}
```

INPUT:

ENTER THE VALUE FOR n
10

OUTPUT:

FIBONACCI SEQUENCE FOR THE FIRST 10 TERMS:

0 1 1 2 3 5 8 13 21 34

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Week: 2

2) a) Write a C program to generate all prime numbers between 1 and n. Where n is the value supplied by the user.

Aim: To print a prime numbers up to 1 to n

Description:

Prime number is a number which is exactly divisible by one and itself only

Ex: 2, 3, 5, 7,;

Algorithm:

Step 1: start

Step 2: read n

Step 3: initialize $i=1, c=0$

Step 4:if $i \leq n$ goto step 5

If not goto step 10

Step 5: initialize $j=1$

Step 6: if $j \leq i$ do the following. If no goto step 7

i)if $i \% j == 0$ increment c

- ii) increment j

iii) goto Step 6

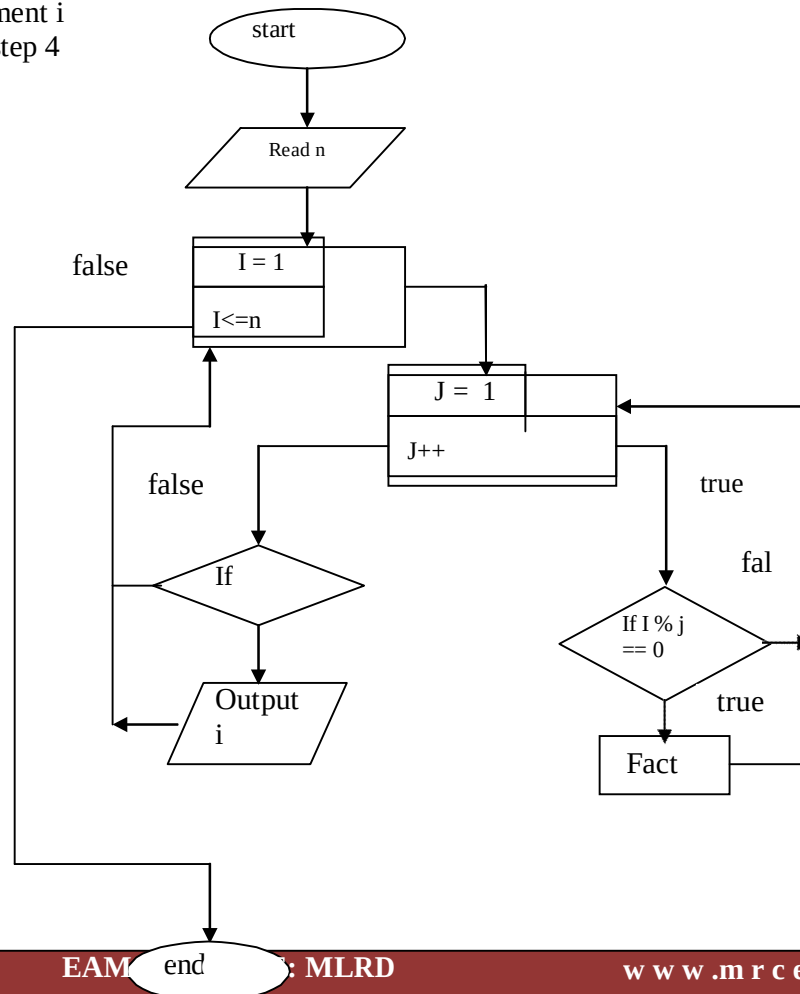
Step 7: if $c == 2$ print i

Step 8: increment i

Step 9: goto step 4

Step 10: stop

FLOWCHART:



Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,i,fact,j;
    clrscr();
    printf("enter the number:");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        fact=0;
        //THIS LOOP WILL CHECK A NO TO BE PRIME NO. OR NOT.
        for(j=1;j<=i;j++)
        {
            if(i%j==0)
                fact++;
        }
        if(fact==2)
            printf("\n %d",i);
    }
    getch( );
}
```

Output:

Enter the number : 5
2 3 5

Record at least 3 results

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2) b) Write a C program to Check whether given number is Armstrong Number or Not.

AIM: To Check whether given number is Armstrong Number or Not

Algorithm:

Armstrong number

Step 1: start

Step 2: read n

Step 3: assign $\text{sum} \leftarrow 0, I \leftarrow m \leftarrow n, \text{count} = 0$

Step 4: if $m > 0$ repeat

Step 4.1: $m \leftarrow m/10$

Step 4.2: $\text{count}++$

Step 4.3: until the condition fail

Step 5: if $I > 0$ repeat step 4 until condition fail

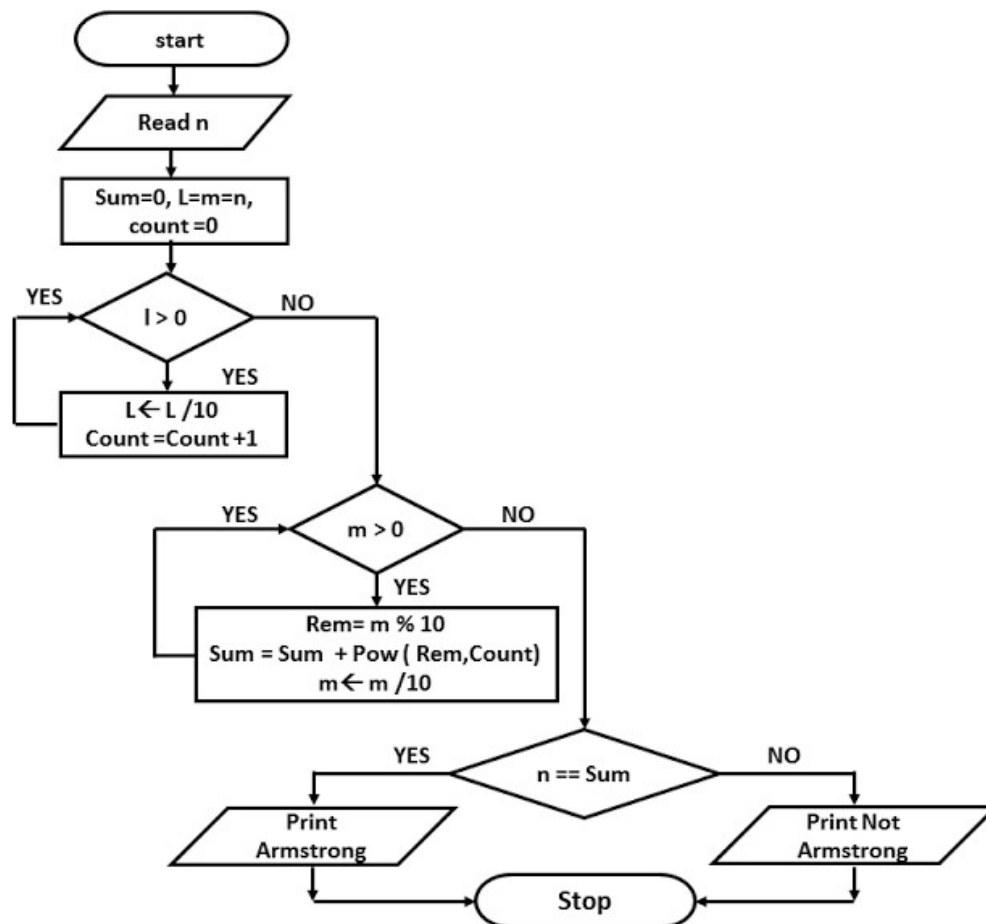
Step 5.1: $\text{rem} \leftarrow I \% 10$

Step 5.2: $\text{sum} \leftarrow \text{sum} + \text{pow}(\text{rem}, \text{count})$

Step 5.3: $I \leftarrow I/10$

Step 6: if $n = \text{sum}$ print Armstrong otherwise print not armstrong

Step 7: stop



Program:

```
#include <stdio.h>
int main()
{
    int n, n1, rem, num=0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    n1=n;
    while(n1!=0)
    {
        rem=n1%10;
        num+=rem*rem*rem;
        n1/=10;
    }
    if(num==n)
        printf("%d is an Armstrong number.",n);
    else
        printf("%d is not an Armstrong number.",n);
}
```

Input:

Enter a positive integer: 371

Output:

371 is an Armstrong number.

Record at least 3 results

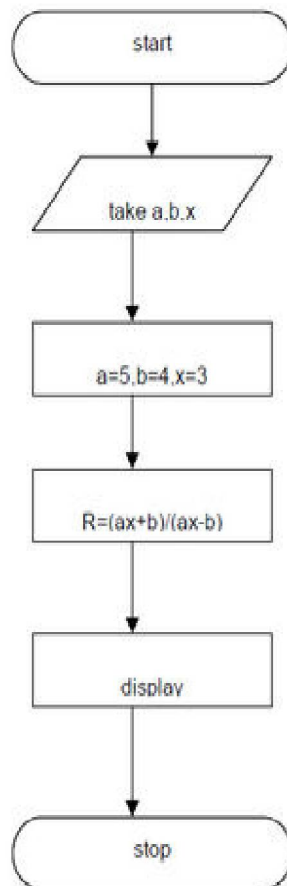
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2) c). Write a C program to evaluate algebraic expression $(ax+b)/(ax-b)$

Algorithm:

Step 1: start
Step 2: input a,b,x,s
Step 3: $s=(a*x+b)/(a*x-b)$
Step 4: Result s
Step 5: stop

Flow Chart:



Program:

```
#include<stdio.h>
#include<conio.h>
int main( )
{
int a,b,x;
float s;
clrscr();
printf("enter the values of a,b,x");
scanf("%d %d %d",&a,&b,&x);
s=(a*x+b)/(a*x-b);
printf("The value of s=%f",s);
getch();
}
```

Input:enter the values of a,b,x

1 3 2

Output:

The value of s= 5

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Week: 3

3) a). Write a C program to check whether given number is perfect number or Not

AIM: To Check whether given number is perfect number or not

Algorithm:**Perfect number**

Step 1: read n

Step 2: assign i=1,sum=0

Step 3: while(i<n) goto step 4

Step 4: if(n%i==0)

sum=sum+i

i++

step 5: if(sum==n) print given number is perfect number otherwise not a perfect number.

Program:

```
#include<stdio.h>
int main()
{
    int n,i=1,sum=0;

    printf("Enter a number: ");
    scanf("%d",&n);

    while(i<n)
    {
        if(n%i==0)
            sum=sum+i;
        i++;
    }
    if(sum==n)
        printf("%d is a perfect number",i);
    else
        printf("%d is not a perfect number",i);

    return 0;
}
```

Input:

Enter a number:6

Output:

6 is a perfect number

3) b) Write a C program to check whether a number is strong number or not.

AIM: To check whether given number is strong number or not

Algorithm:

Strong number

Step 1: read num, i, f, r, sum=0, temp

Step 2: assign num to temp

Step 3: while(num) goto step 4

Step 4: i=1, f=1

 r=num%10

 while(i<=r) goto step 5

Step 5: f=f*i

 i=i+1

Step 6: sum=sum+f;

Step 7: num=num/10;

Step 8: if sum and temp are equal got step 9

Step 9: print strong number otherwise not a strong number

Program:

```
#include<stdio.h>
```

```
int main() {
```

```
    int num,i,f,r,sum=0,temp;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d",&num);
```

```
    temp=num;
```

```
    while(num) {
```

```
        i=1,f=1;
```

```
        r=num%10;
```

```
        while(i<=r) {
```

```
            f=f*i;
```

```
            i++;
```

```
        }
```

```
        sum=sum+f;
```

```
        num=num/10;
```

```
    }
```

```
    if(sum==temp)
```

```
        printf("%d is a strong number",temp); else
```

```
        printf("%d is not a strong number",temp);
```

```
    return 0;
```

```
}
```

Input:

Enter a number:145

Output:

145 is a strong number

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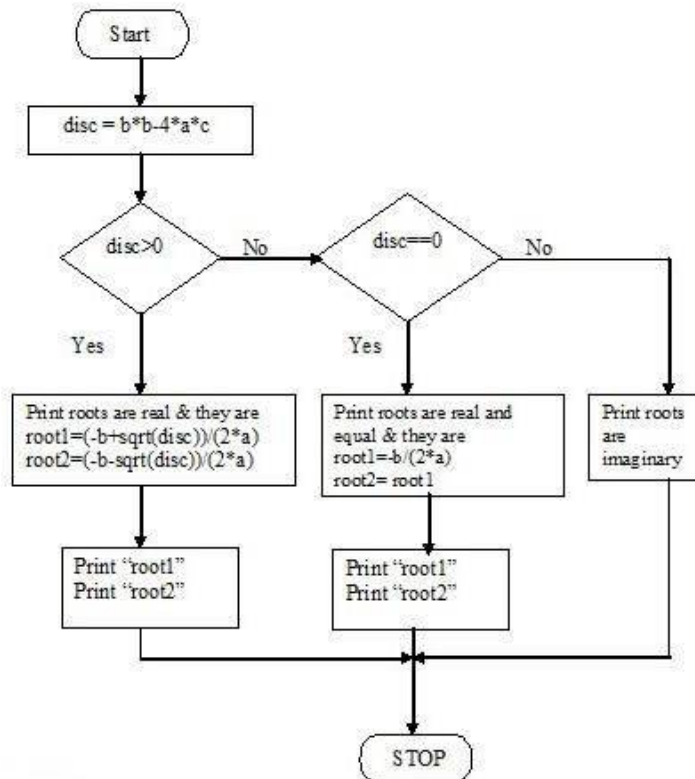
Week: 4**4) a) Write a C program to find the roots of a quadratic equation.**

AIM: To find the roots of a quadratic equation.

Description: roots of quadratic equation are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

ALGORITHM:

Step 1: Start
 Step 2: Read a,b,c
 Step 3: calculate disc = $b*b-4*a*c$
 Step 4: if(disc>0)
 Begin
 Step 5: root1= $(-b+\sqrt{\text{disc}})/(2*a)$
 Step 6: root2= $(-b-\sqrt{\text{disc}})/(2*a)$
 Step 7: Print "Root1" , "Root2"
 End
 Step 8: else if(disc=0)
 Begin
 Step 9: root1= $-b/(2*a)$
 Step 10: root2=root1;
 Step 11: Print "Root1" , "Root2"
 End
 Step 12: else
 Step 13: Print Roots are imaginary
 Step 14: Stop

Flow Chart

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
int main()
{
    int a,b,c;
    float disc,root1,root2;
    float img,real;
    printf("ENTER VALUES FOR a,b,c:\n");
    scanf("%d%d%d",&a,&b,&c);
    disc=(float)b*b-4*a*c;
    if(disc>0)
    {
        printf("THE ROOTS ARE REAL & UNEQUAL:\n");
        root1=(-b+sqrt(disc))/(2*a);
        root2=(-b-sqrt(disc))/(2*a);
        printf("Root1=%f\n",root1);
        printf("Root2=%f\n",root2);
    }
    else if(disc==0)
    {
        printf("THE ROOTS ARE REAL AND EQUAL:\n");
        root1=-b/(2*a);
        root2=root1;
        printf("Root1=%f\n",root1);
        printf("Root2=%f\n",root2);
    }
    else
    {
        printf("THE ROOTS ARE IMAGINARY:\n");
        disc=-disc;
        img=(float)disc/2*a;
        real=(float)-b/2*a;
        if (img>0)
        {
            printf("Root1=%f + i%f\n",real,img);
            printf("Root2=%f - i%f\n",real,img);
        }
        else
        {
            img=-img;
            printf("Root1=%f + i%f\n",real,img);
            printf("Root2=%f - i%f\n",real,img);
        }
    }
    return 0;
}
```

INPUT:

ENTER VALUES FOR a,b,c

1 4 4

OUTPUT:

THE ROOTS ARE EQUAL AND THEY ARE.. Root1=-2 Root2=-2

Record at least 3 results**Signature of faculty with date**

4) b). Write a C program which takes two integer operands and one operator from the user, performs the operation and then prints the result.(Consider the operators +,-,*,/,% and use Switch Statement.)

AIM:

To perform arithmetic operations using switch statement.

Algorithm:

Step 1: Read a,b

Step 2: Print “Menu Options”

Step 3: do

Begin Step 4: Read ch

Step 5: switch(ch)

Begin Step 6:

case 1:

Begin

Calculate $c = a + b$

Print “c”

break;

End

case 2:

Begin

Calculate $c = a - b$

Print “c”

break;

End

case 3:

Begin

Calculate $c = a * b$

Print “c”

break;

End

case 4:

Begin

Calculate $c = a / b$

Print “c”

break;

End

case 5:

Begin

Calculate $c = a \% b$

Print “c”

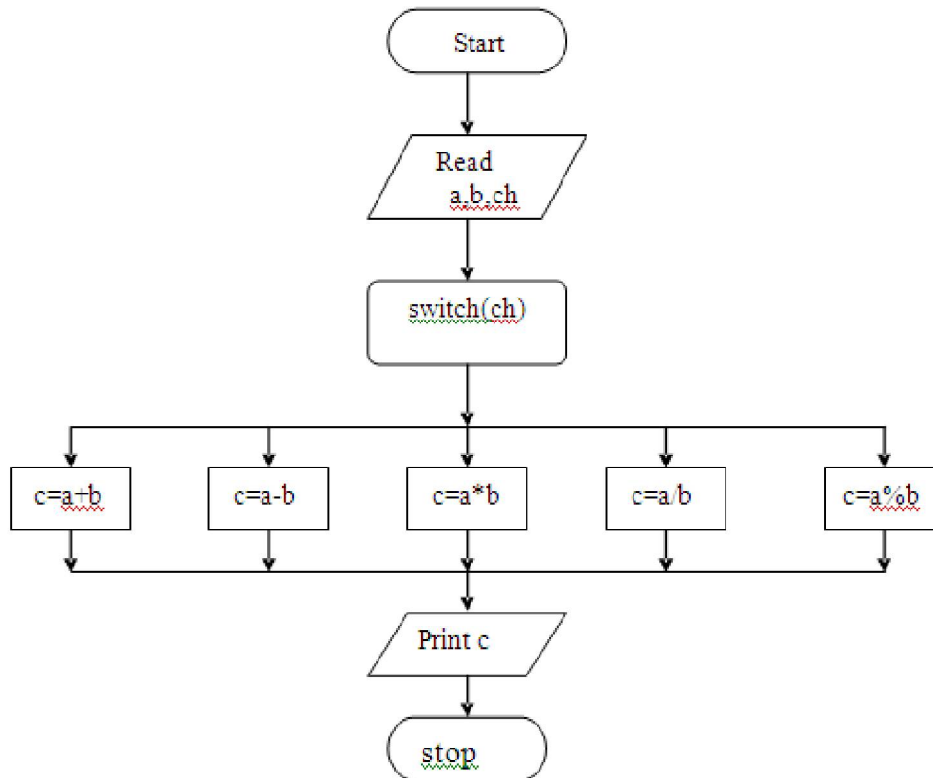
break;

End

default:

Print “Invalid choice”

End

Flowchart**Program:**

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,c,ch; clrscr();
    printf("ENTER TWO VALUES FOR a & b\n"); scanf("%d %d",&a,&b);
    while(1) {
        printf("MENU OPTIONS \n");
        printf("*****\n");
        printf("1.Addition\n");
        printf("2.Subtraction\n");
        printf("3.Multiplication\n");
        printf("4.Division\n");
        printf("5.Modulus\n");
        printf("6.Exit\n");
        printf("\n");
        printf("ENTER UR CHOICE\n");
        scanf("%d",&ch);
        switch(ch) {
```

```
case 1: c=a+b;
printf("The addition of %d and %d is..%d\n",a,b,c); break;
case 2: c=a-b;
printf("The subtraction of %d and %d is..%d\n",a,b,c); break;
case 3: c=a*b;
printf("The multiplication of %d and %d is..%d\n",a,b,c); break;
case 4: c=a/b;
printf("The division of %d and %d is..%d\n",a,b,c); break;
case 5: c=a%b;
printf("The modulus of %d and %d is..%d\n",a,b,c); break;
case 6:exit(0); default:printf("INVALID CHOICE\n"); }
}
getch();
}
```

INPUT:

ENTER TWO VALUES FOR a & b: 20 16

OUTPUT:

MENU OPTIONS

1.Addition 2.Subtraction 3.Multiplication 4.Division 5.Modulus
6.Exit

ENTER UR CHOICE 1

The addition of 20 and 16 is..36

Record at least 3 results

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Week: 5

5) a) Write a C program to find the factorial of a given integer using non-recursive function.

AIM:

To find the factorial of a given number using non-recursive function.

ALGORITHM:

Step 1: Start

Step 2: Read n

Step 3: Call fact(n) goto step 6

Step 4: Store result in “f”

Step 5: Print “f” goto step 10

Step 6: Begin //sub program

Initialize $f \leftarrow 1$

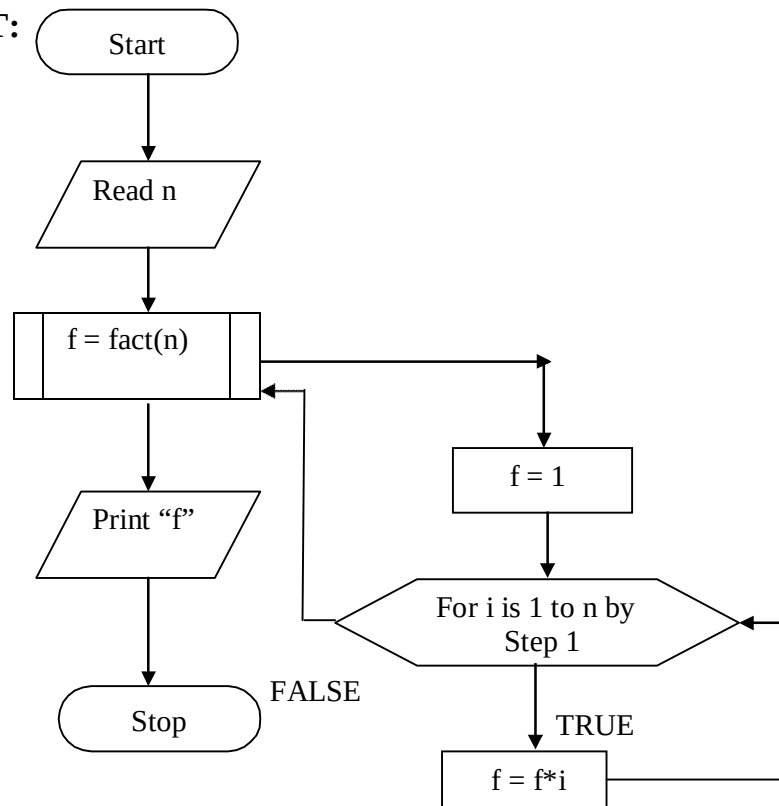
Step 7: for i is 1 to n by step 2

Step 8: Calculate $f = f*i$

Step 9: return “f”

End

Step 10: Stop

FLOWCHART:

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
int fact(int);
void main()
{
    int n,i,f;
    clrscr();
    printf("ENTER A VALUE FOR n:\n");
    scanf("%d",&n);
    f=fact(n);
    printf("THE FACTORIAL OF A GIVEN NO IS..%d",f);
    getch();
}
int fact(int n)
{
    int i,f=1;
    for(i=1;i<=n;i++)
        f=f*i;
    return(f);
}
```

INPUT:

ENTER A VALUE FOR n

5

OUTPUT:

THE FACTORIAL OF A GIVEN NUMBER IS..120

Record at least 3 results

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5) b) Write a C program to find the factorial of a given integer using recursive function.

AIM:

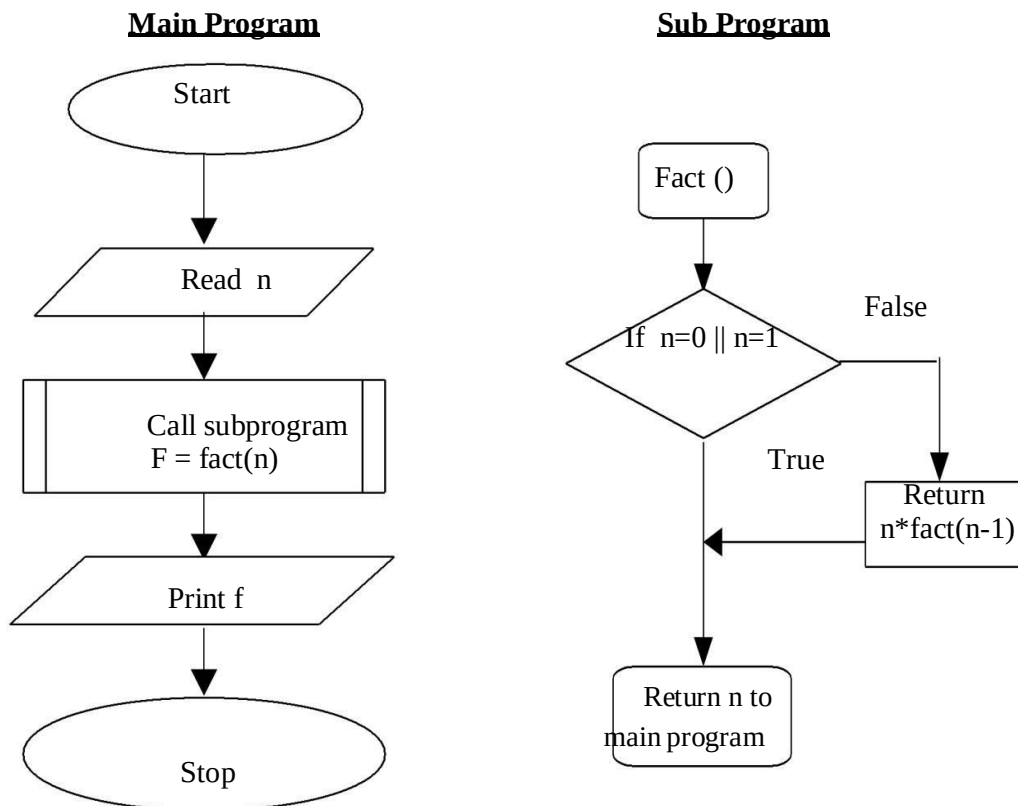
To find the factorial of a given number using recursive function.

ALGORITHM:**main program**

Step 1: start
Step 2: read n
Step 3: call sub program as $f = \text{fact}(n)$
Step 4: print f value
Step 5: stop

Sub program:

Step 1: initialize the f
Step 2: if $n = 0$ or $n == 1$ return 1 to main program if not goto step 3
Step 3: return $n * \text{fact}(n-1)$ to main program

FLOW CHART:

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
int fact(int);
void main()
{
    int n,res;
    clrscr();
    printf("ENTER A NUMBER:\n");
    scanf("%d",&n);
    res=fact(n);
    printf("THE FACTORIAL OF A GIVEN NUMBER IS..%d",res);
    getch();
}
int fact(int n)
{
    int r;
    if(n==0)
        return(1);
    else
    {
        r=n*fact(n-1);
        return(r);
    }
}
```

INPUT:

ENTER A VALUE FOR n
5

OUTPUT:

THE FACTORIAL OF A GIVEN NUMBER IS..120

Record at least 3 results

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Week: 6

6) a) Write a C program to find the GCD of two given integers by using the recursive function

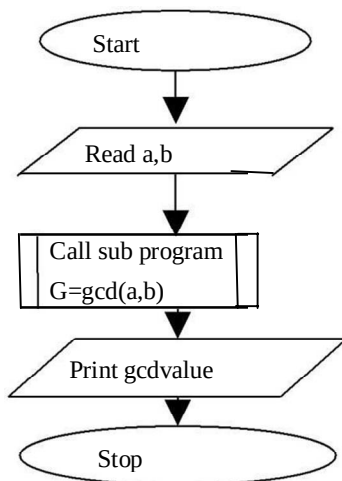
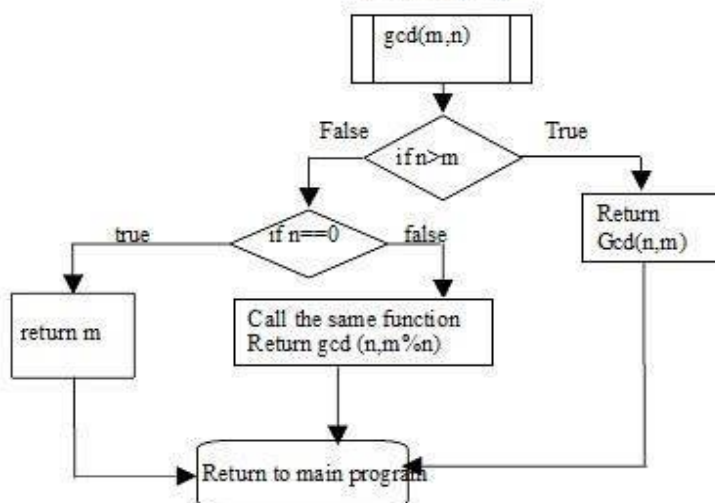
Aim: To find the Gcd of two given integers by using the recursive function

Algorithm:**Main program:**

Step 1: start
 Step 2: read a,b
 Step 3: call the sub program GCD(a,b) for print the value
 Step 4: stop

Sub program: GCD(n,m)

Step 1: if $n > m$ return GCD(n,m)
 Step 2: if $n == 0$ return m else goto step 3
 Step 3: return GCD (n,m%n)
 Step 4: return to main program

Flow CHART:**Main Program:****Sub Program:**

Program:

```
#include<stdio.h>
#include<conio.h>
int gcdrecursive(int m,int n)
{
    if(n>m)
        return gcdrecursive(n,m);
    if(n==0)
        return m;
    else
        return gcdrecursive(n,m%n);    // return to the main program
}
void main()
{
    int a,b,igcd; clrscr();
    printf("enter the two numbers whose gcd is to be found:");
    scanf("%d%d",&a,&b);
    printf("GCD of a,b is %d",gcdrecursive(a,b)); // return to the sub program getch();
}
```

Input:

Enter the two numbers whose gcd is to be found: 5 25

Output:

GCD of a,b is : 5

Record at least 3 results

Signature of faculty with date

6) b) Write a C program to find the GCD of two given integers using non-recursive function.

To find the GCD of two given integers by using the non recursive function

Description:

GCD means Greatest Common Divisor. i.e the highest number which divides the given number

Ex: GCD(12,24) is 12

Formula: $\text{GCD} = \frac{\text{product of numbers}}{\text{LCM of numbers}}$

Algorithm:

Step 1: start

Step 2: read a,b

Step 3: call sub program $g = \text{GCD}(a,b)$

Step 4: print the g value

Step 5: stop

Sub program:

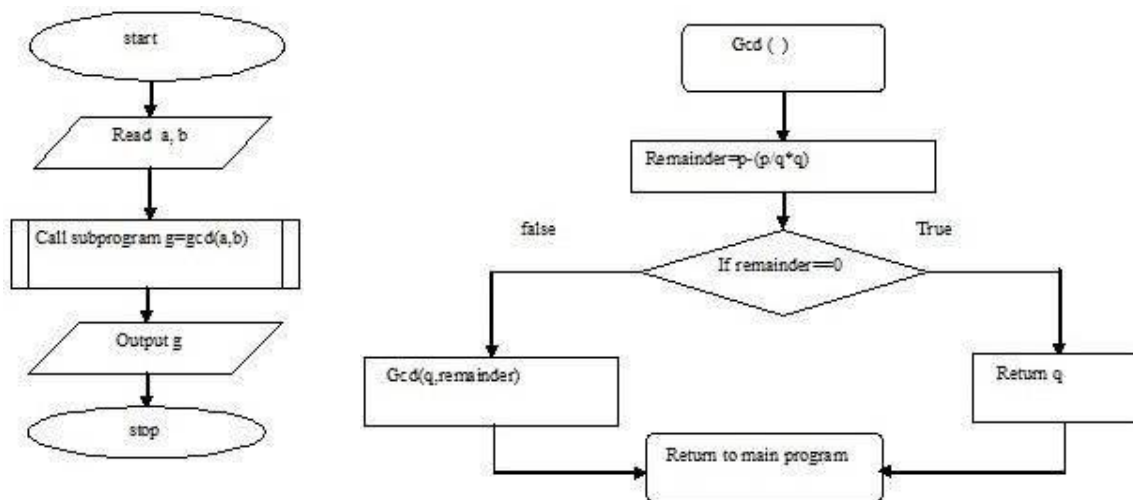
Step 1: initialize the $p=1$, q, remainder

Step 2: $\text{remainder} = p - (p/q * q)$

Step 3: $\text{remainder} = 0$ return q else goto step 4

Step 4: $\text{GCD}(q, \text{remainder})$ return to main program

Flowchart:



Program:

```

#include<stdio.h>
#include<conio.h>
#include<math.h>
int gcdnonrecursive(int m,int n)
{
    int remainder;

```

```
        remainder=m-(m/n*n);
        if(remainder==0)
            return n;
        else
            gcdnonrecursive(n,remainder);
    }

void main()
{
    int a,b,igcd;
    clrscr();
    printf("enter the two numbers whose gcd is to be found:");
    scanf("%d%d",&a,&b);
    printf("GCD of %d",gcdnonrecursive(a,b));
    getch();
}
```

Output:

1. enter the two numbers whose gcd is to be found:5,25
GCD of a,b is : 5

Record at least 3 results

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Week: 7

7) a) Write a C program to find both the largest and smallest number in a list of integers.

AIM:

To find the largest and smallest number in a list of integers.

ALGORITHM:

Step 1: start

Step 2: read n

Step 3: initialize i=0

Step 4: if i<n do as follows. If not goto step 5

Read a[i]

Increment i

Goto step 4

Step 5: small=a[0], large=a[0]

Step 6: initialize i=0

Step 7: if i<n do as follows. If

not goto step 8

If a[i]<small

Assign small=a[i]

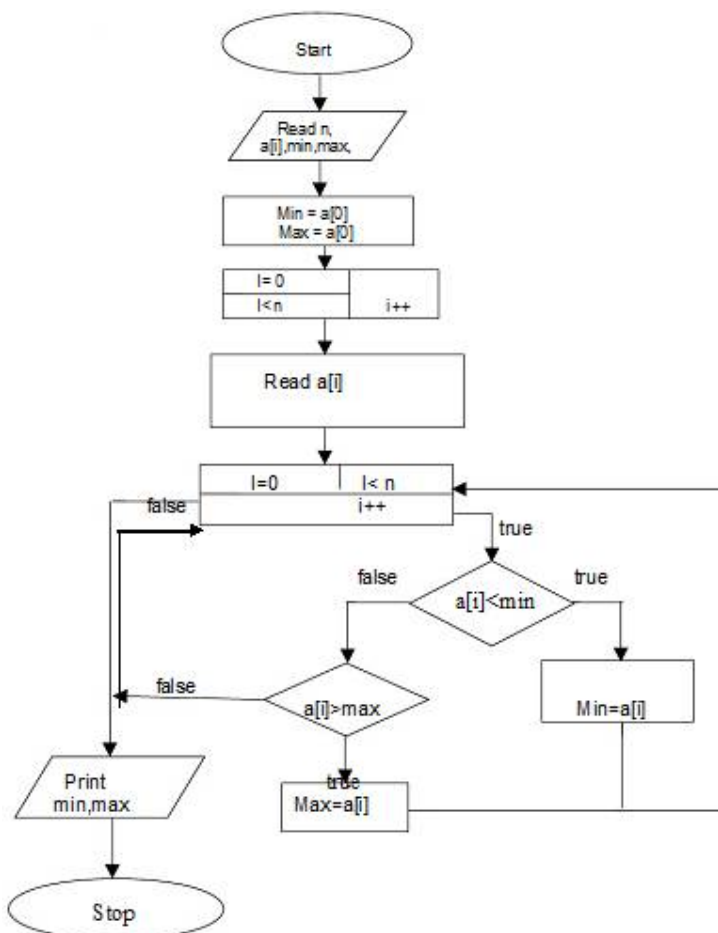
If a[i]>large

Assign large=a[i]

Increment i goto Step 7

Step 8: print small, large

Step 9: stop



Program:

```
#include<stdio.h>
#include<conio.h>
void main()
{ int a[10],i,n,small,large;
  clrscr();
  printf("Enter The Array Size:");
  scanf("%d",&n);
  printf("ENTER ELEMENTS OF ARRAY");
  for(i=0;i<n;i++) // read the elements of an array
  scanf("%d",&a[i]);
  small=a[0];
  large=a[0];
  for(i=0;i<n;i++)// read the elements of an array
  {   if(a[i]<small)// check the condition for minimum value
      small=a[i];
      if(a[i]>large)//check the condition for maximum value
      large=a[i];
  }
  printf("largest value is:%d\n",large);
  printf("smallest value is:%d\n",small);
  getch();
}
```

INPUT:

Enter The Array Size:10

ENTER THE ELEMENTS OF ARRAY

7 10 9 8 6 5 2 3 4 1

OUTPUT:

largest value is : 10

smallest value is : 1

Record at least 3 results

Signature of faculty with date

7) b) Write a C Program to Sort the Array in an Ascending Order.**Program:**

C Program to Sort the Array in an Ascending Order

```
#include <stdio.h>
```

```
void main()
```

```
{
```

```
    int i, j, a, n, number[30];
```

```
    printf("Enter the value of N \n");
```

```
    scanf("%d", &n);
```

```
    printf("Enter the numbers \n");
```

```
    for (i = 0; i < n; ++i)
```

```
        scanf("%d", &number[i]);
```

```
    for (i = 0; i < n; ++i)
```

```
    {
```

```
        for (j = i + 1; j < n; ++j)
```

```
        {
```

```
            if (number[i] > number[j])
```

```
            {
```

```
                a = number[i];
```

```
                number[i] = number[j];
```

```
                number[j] = a;
```

```
            }
```

```
        }
```

```
    }
```



```
printf("The numbers arranged in ascending order are given below \n");
```

```
for (i = 0; i < n; ++i)
```

```
    printf("%d\n", number[i]);
```

```
}
```

Output:

Enter the value of N

6

Enter the numbers

3

78

90

456

780

200

The numbers arranged in ascending order are given below

3

78

90

200

456

780

Record at least 3 results

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7) c) Write a C Program to find whether given matrix is symmetric or not.**Program:**

```
#include<conio.h>
#include<stdio.h>
void main()
{
    int a[10][10],i,j,m;
    clrscr();
    printf("Enter order of square matrix: ");
    scanf("%d",&m);
    for(i=0;i<m;i++)
    {
        for(j=0;j<m;j++)
        {
            printf("Enter value of a[%d][%d]: ",i,j);
            scanf("%d",&a[i][j]);
        }
    }
    for(i=0;i<m;i++)
    {
        for(j=0;j<m;j++)
        {
            if(a[i][j]!=a[j][i])
            {
                printf("\n\nMatrix is not symmetric");
                getch();
                exit(0);
            }
        }
    }
    printf("\n\nMatrix is symmetric");
    getch();
}
```

Output:

```
Enter order of square matrix:2
Enter value of a[0][0]:12
Enter value of a[0][1]:34
Enter value of a[1][0]:34
Enter value of a[1][1]:54
Matrix is symmetric
```

Week : 9**9) a) Write a C program to perform addition of two matrices.****AIM:**

To perform addition of two matrices.

ALGORITHM:

Step 1: Start

Step 2: for i is 0 to 2 by step 1

for j is 0 to 2 by step 1

Step 3: Read $a[i][j], b[i][j]$

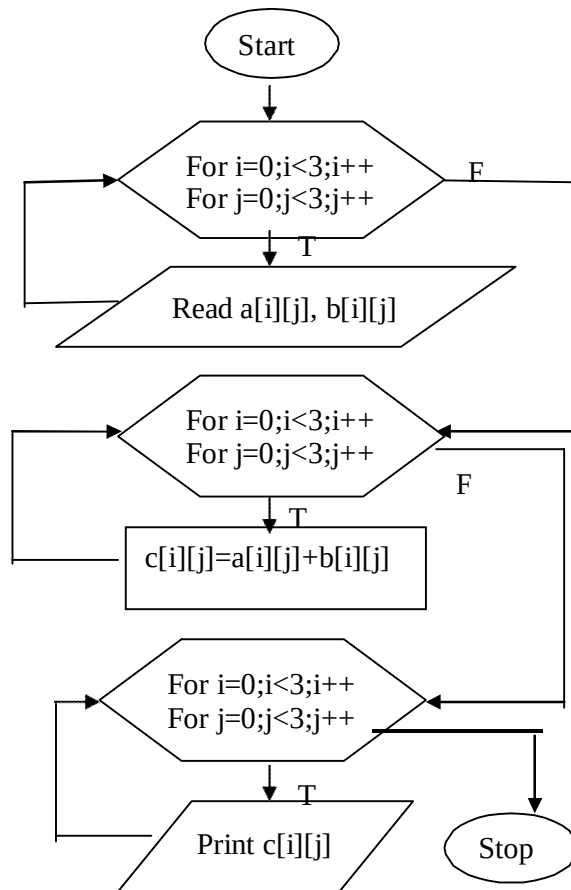
Step 4: goto step 2

Step 5: calculate $c[i][j] = a[i][j] + b[i][j]$

Step 6: goto step 2

Step 7: Print $c[i][j]$

Step 8: Stop

Flow Chart:

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3][3],b[3][3],c[3][3];
int i,j;
    clrscr();
    printf("ENTER A MATRIX\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
            scanf("%d",&a[i][j]);
    }
    printf("ENTER B MATRIX\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
            scanf("%d",&b[i][j]);
    }
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
            c[i][j]=a[i][j]+b[i][j];
    }
    printf(" After addition of two matrices :\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("%d\t",c[i][j]);
        }
        printf("\n");
    }
    getch();
}
```

INPUT:

ENTER a MATRIX

1 2 3

4 5 6

7 8 9

ENTER b MATRIX

1 1 1

1 1 1

1 1 1

OUTPUT:

After addition of two matrices is..

2 3 4

5 6 7

8 9 10

Record at least 3 results**Signature of faculty with date**

9) b) Write a C program that uses functions to perform Multiplication of Two Matrices.**AIM:**

To perform multiplication of two matrices.

ALGORITHM:

Step 1: Start

Step 2: for i is 0 to 2 by step 1

for j is 0 to 2 by step 1

Step 3: Read a[i][j], b[i][j]

Step 4: goto step 2

Step 5: calculate $c[i][j] = c[i][j] + a[i][k] * b[k][j]$

Step 6: goto step 2

Step 7: Print c[i][j]

Step 8: Stop

Program:

```
#include<stdio.h >
#include<conio.h>
int i,j,k;
void main()
{
    int a[10][10],b[10][10],c[10][10],m,n,p,q;
    void mul(int x[10][10],int y[10][10],int z[10][10],int m,int n,int p,int q);
    void read(int x[10][10],int m,int n);
    void display(int x[10][10], int m,int n);
    clrscr();
    printf("Enter the size of A Mtrix (Row and Col): \n");
    scanf("%d%d",&m,&n);
    printf("Enter the size of B Mtrix (Row and Col): \n");
    scanf("%d%d",&p,&q);
    if(n!=p)
    {
        printf("Multiplication Not Possible\n Please re-enter\n");
        printf("correct size and try again..... \n");
    }
    else
    {
        read(a,m,n);
        read(b,m,n);
        mul(a,b,c,m,n,p,q);
        printf("A Matrix is :\n");
        display(a,m,n);
    }
}
```

```
printf("B Matrix is :\n");

display(b,m,n);
printf("C Matrix is :\n");
display(c,m,n);
}
getch();
}
void mul(int x[10][10],int y[10][10],int z[10][10],int m,int n,int p,int q)
{
    for (i=0;i<m;i++)
        for(j=0;j<q;j++)
        {
            z[i][j]=0;
            for(k=0;k<n;k++)
                z[i][j]+= x[i][k]*y[k][j];
        }
}

void read(int x[10][10], int m,int n)
{
    printf("Enter Matrix Value Row by Row\n");
    for (i=0;i<m;i++)
        for(j=0;j<n;j++)
            scanf("%d",&x[i][j]);
}

void display(int x[10][10], int m,int n)
{
    for (i=0;i<m;i++)
    {
        for(j=0;j<n;j++)
            printf("%5d",x[i][j]);
        printf("\n");
    }
    printf("\n");
}
```

Input:

Enter the size of A Mtrix (Row and Col): 2 2

Enter the size of B Mtrix (Row and Col): 2 2

Enter Matrix Value Row by Row

1 0
2 6

Enter Matrix Value Row by Row

3 4
4 2

Output:

A matrix is:

1 0
2 6

B Matrix is:

3 4
4 2

C matrix is:

3 4
24 20

Record at least 3 results

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Algorithm:

Step 1: start
Step 2: read main string and sub string
Step 3: find the length of main string(r)
Step 4: find length of sub string(n)
Step 5: copy main string into sub string
Step 6: read the position to insert the sub string(p)
Step 7: copy sub string into main string from position p-1
Step 8: copy temporary string into main string from position p+n-1
Step 9: print the strings
Step 10: stop

Program:

```
#include<stdio.h>
#include<string.h>
main()
{
char a[30],b[30],c[30];
int pos=0,i=0,l,la,lb,lc,j;
    puts("Enter a string");
    gets(a);
    puts("Enter sub string");
    gets(b);
    puts("enter position for insertion");
    scanf("%d",&pos);
    la=strlen(a);
    lb=strlen(b);
    l=pos+lb;
    lc=la+lb;
    for(i=0;i<pos;i++)
    {
        c[i]=a[i];
    }
    j=0;
    for(i=pos;i<=l;i++)
    {
        c[i]=b[j];
        j++;
    }
    j=pos;
    for(i=l;i<lc;i++)
    {
        c[i]=a[j];
        j++;
    }
    c[i]='\0';
    puts("String after Insertion is:");
    printf("%s",c);
}
```

```
}
```

Input:

Enter First String:

Comer

Enter Second String:

put

Output:

Enter the position where the item has to be inserted:3

Computer

Record at least 3 results

Signature of faculty with date

10) b) To delete n Characters from a given position in a given string.

Aim: To delete n Characters from a given position in a given string.

Algorithm:

Step 1: start

Step 2: read string

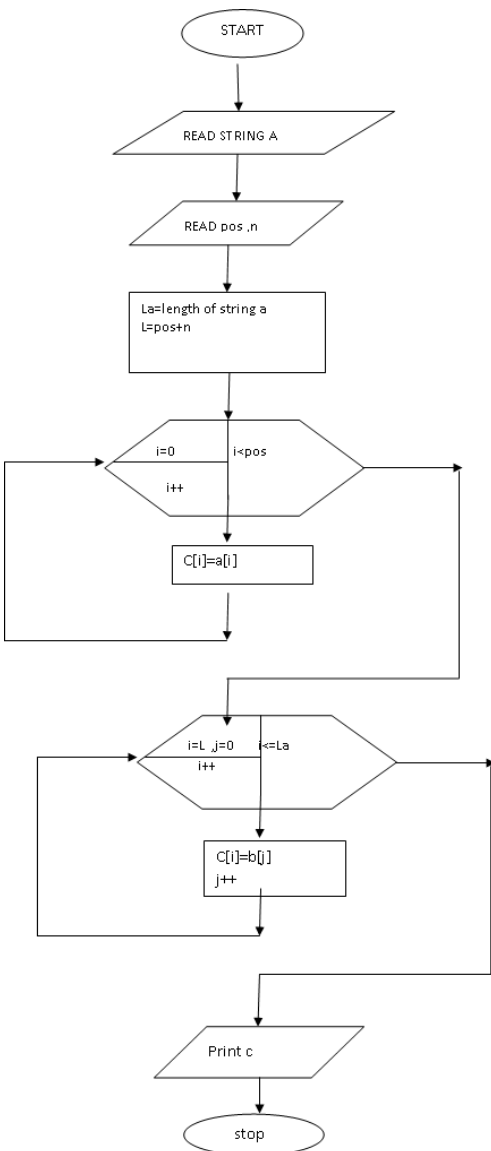
Step 3: find the length of the string

Step 4: read the value of number of characters to be deleted and positioned

Step 5: string copy part of string from position to end, and
(position + number of characters to end)

Step 6: stop

Flow Chart:



Program:

```
#include<stdio.h>
#include<string.h>
main()
{
    char a[30],c[30];
    int pos=0,i=0,L,La,j,n;
    puts("Enter a string");
    gets(a);
    puts("enter position for deletion");
    scanf("%d",&pos);
    puts("Enter number of characters to be deleted");
    scanf("%d",&n);
    La=strlen(a);
    L=pos+n;
    for(i=0;i<pos;i++)
    {
        c[i]=a[i];
    }
    j=pos;
    for(i=L;i<=La;i++)
    {
        c[j]=a[i];
        j++;
    }
    puts("String after Deletion is:");
    printf("%s",c);
}
```

Input:

Enter the string
jayapal

Enter the position from where to delete:4
Enter the number of characters to be deleted 2

Output:

jayal

Record at least 3 results

Signature of faculty with date

Week: 11

11) a) Write a C program using user defined functions to determine whether the given string is palindrome or not.

Aim: To determine if the given string is palindrome or not.

Description :

Palindrome means string on reversal should be same as original

Ex: madam on reversal is also madam

Algorithm:

Step 1: start

Step 2: read string A

Step 3: copy string A into B

Step 4: reverse string B

Step 5: compare A & B

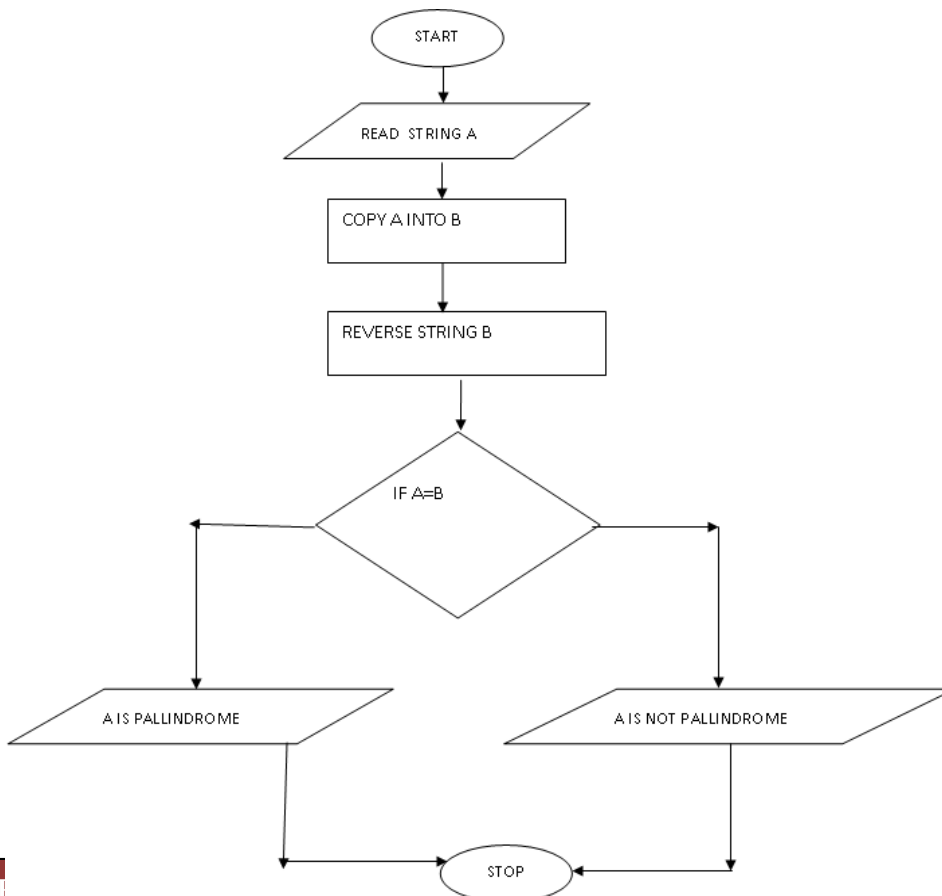
If A equals B to go to step 6

Else goto step 7

Step 6: print given string A is pallindrom

Step 7: print given string is not pallindroma

Step 8: stop

Flow Chart:

Program:

```
#include <stdio.h>
#include <string.h>
void main()
{
    char string[25], reverse_string[25] = {'\0'};
    int i, length = 0, flag = 0;

    printf("Enter a string \n");
    gets(string);
    for (i = 0; string[i] != '\0'; i++)
    {
        length++;
    }
    printf("The length of the string '%s' = %d\n", string, length);
    for (i = length - 1; i >= 0 ; i--)
    {
        reverse_string[length - i - 1] = string[i];
    }

    for (flag = 1, i = 0; i < length ; i++)
    {
        if (reverse_string[i] != string[i])
            flag = 0;
    }
    if (flag == 1)
        printf ("%s is a palindrome \n", string);
    else
        printf ("%s is not a palindrome \n", string);
}
```

Input:

Enter a string
madam

Output:

The length of the string 'madam' = 5
madam is a palindrome

Record at least 3 results

Signature of faculty with date

11. b) Write a C program that displays the position or index in the string S where the string T begins, or - 1 if S doesn't contain T.

Aim: To display the position or index in the string S where the string T begins, or - 1 if S doesn't contain T

Algorithm:

Step 1: start

Step 2: read the string and then displayed

Step 3: read the string to be searched and then displayed

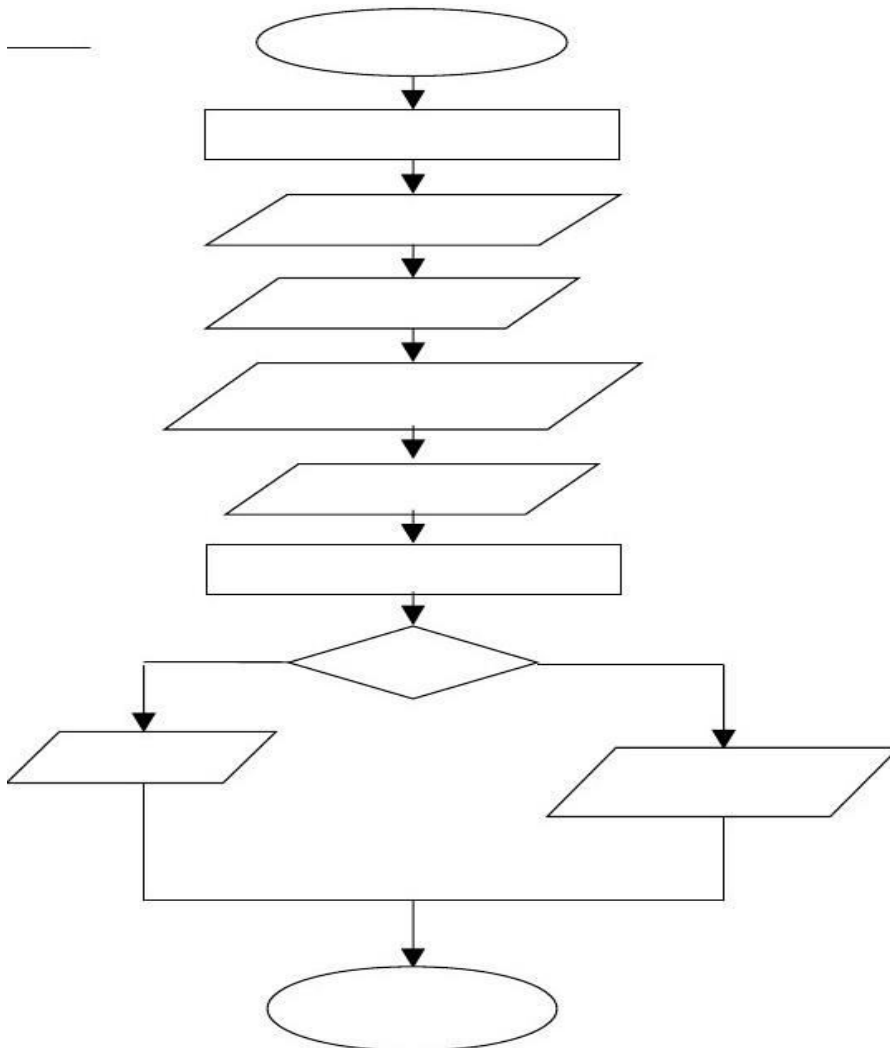
Step 4: searching the string T in string S and then perform the following steps

i. found=strstr(S,T)

ii. if found print the second string is found in the first string at the position. If not goto step 5

Step 5: print the -1

Step 6: stop

Flow Chart:

Program:

```
#include<stdio.h>
#include<string.h>
#include<conio.h>
void main()
{
    char s[30], t[20];
    char *found;
    clrscr();
    /* Entering the main string */
    puts("Enter the first string: ");
    gets(s);
    /* Entering the string whose position or index to be displayed */
    puts("Enter the string to be searched: ");
    gets(t);
    /*Searching string t in string s */
    found=strstr(s,t);
    if(found)
        printf("Second String is found in the First String at %d position.\n",found-s);
    else
        printf("-1");
    getch();
}
```

Input:

Enter the first string:
computer
Enter the string to be seareched:
mp

Output:

Second string is found in the first string at 2 position

Record at least 3 results

Signature of faculty with date

Week: 12

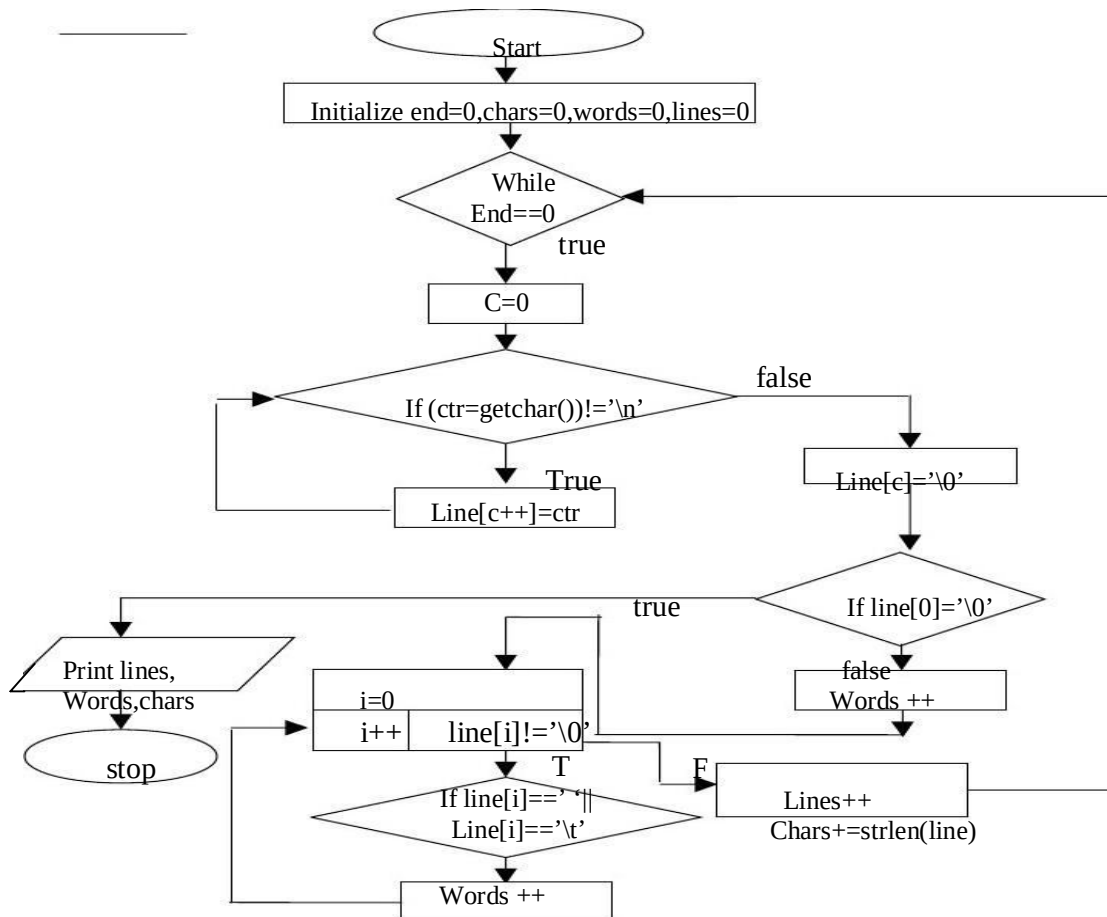
12) a) Write a C program to count the number of lines, words and characters in a given text.

AIM:

To count the number of lines, words and characters in a given list.

ALGORITHM:

- Step 1: Start
 Step 2: Read the text until an empty line
 Step 3: Compare each character with newline char '\n' to count no of lines
 Step 4: Compare each character with tab char '\t' or space char ' ' to count no of words
 Step 5: Compare first character with NULL char '\0' to find the end of text
 Step 6: No of characters = length of each line of text
 Step 7: Print no of lines, no of words, no of chars
 Step 8: Stop.

Flow Chart:

PROGRAM:

```
#include <stdio.h>
void main()
{
    char line[81], ctr;
    int i,c,
        end = 0,
        characters = 0,
        words = 0,
        lines = 0;
    printf("TYPE ANY TEXT.\n");
    printf("GIVE ONE SPACE AFTER EACH WORD.\n");
    while( end == 0)
    {
        /* Reading a line of text */
        c = 0;
        while((ctr=getchar()) != '\n')
            line[c++] = ctr;
        line[c] = '\0';
        /* counting the words in a line */
        if(line[0] == '\0')
            break ;
        else
        {
            words++;
            for(i=0; line[i] != '\0';i++)
                if(line[i] == ' ' || line[i] == '\t')
                    words++;
        }
        /* counting lines and characters */
        lines = lines +1;
        characters = characters + strlen(line);
    }
    printf ("\n");
    printf("Number of lines = %d\n", lines);
    printf("Number of words = %d\n", words);
    printf("Number of characters = %d\n", characters);
}
```

INPUT:

TYPE ANY TEXT

GIVE ONE SPACE AFTER EACH WORD.

Ramu is a good boy.

OUTPUT:

THE NUMBER OF CHARACTERS IN A GIVEN TEXT IS..18

THE NUMBER OF WORDS IN A GIVEN TEXT IS..5

THE NUMBER OF LINES IN A GIVEN TEXT IS..1

Record at least 3 results

Signature of faculty with date

12) b) Write a C program to find the length of the string using Pointer.

Program:

```
#include<stdio.h>
#include<conio.h>

int string_ln(char*);

void main() {
    char str[20];
    int length;
    clrscr();

    printf("\nEnter any string : ");
    gets(str);

    length = string_ln(str);
    printf("The length of the given string %s is : %d", str, length);
    getch();
}

int string_ln(char*p) /* p=&str[0] */
{
    int count = 0;
    while (*p != '\0') {
        count++;
        p++;
    }
    return count;
}
```

Input:

Enter the String : pritesh

Output:

Length of the given string pritesh is : 7

Record at least 3 results

Signature of faculty with Date

Week: 13

13) a) Write a C program to Display array elements using calloc() function.

AIM:

To write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc() function.

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i, n;
    int *a;

    printf("Number of elements to be entered:");
    scanf("%d",&n);

    a = (int*)calloc(n, sizeof(int));
    printf("Enter %d numbers:\n",n);
    for( i=0 ; i < n ; i++ )
    {
        scanf("%d",&a[i]);
    }

    printf("The numbers entered are: ");
    for( i=0 ; i < n ; i++ )
    {
        printf("%d ",a[i]);
    }

    return(0);
}
```

Output:

```
Number of elements to be entered:3
Enter 3 numbers:
22
55
14
The numbers entered are: 22 55 14
```

Record at least 3 results

Signature of faculty with Date

13) b) Write a C Program to Calculate Total and Percentage marks of a student using structure.

Program:

```
#include<stdio.h>
#include<conio.h>
struct student
{
    int rl;
    char nm[20];
    int m1;
    int m2;
    int m3;
    int t;
    float per;
};
void main()
{
    struct student a;
    clrscr();
    printf(" Enter RollNo, Name amd three sub marks\n");
    scanf("%d%s%d%d%d",&a.rl,&a.nm,&a.m1,&a.m2,&a.m3);
    a.t=a.m1+a.m2+a.m3;
    a.per=a.t/3.0;
    printf("rollno=%d\n",a.rl);
    printf("Name=%sk\n",a.nm);
    printf("m1=%d\n",a.m1);
    printf("m2=%d\n",a.m2);
    printf("m3=%d\n",a.m3);
    printf("total=%d\n",a.t);
    printf("per=%f\n",a.per);
    getch();
}
```

Input:

Enter RollNo, Name and three sub marks

12 rama 30 40 50

Output:

rollno=12

Name=rama

m1=30

m2=40

m3=50

total=120

per=40.000000

Week: 14

14) a) Write a C program that uses functions to perform the following operations:

- i) Reading a complex number**
- ii) Writing a complex number**
- iii) Addition of two complex numbers**
- iv) Multiplication of two complex numbers**

AIM: To perform arithmetic operations on complex numbers

Complex numbers of type $a+ib$

Addition: $(a+ib)+(x+iy)=a+x+i(b+y)$

Subtraction: $(a+ib)-(x+iy)=a-x+i(b-y)$

Multiplication: $(a+ib)*(x+iy)=ax-by+i(ay+bx)$

Division

$$(a+ib)/(x+iy) = \frac{a+b}{x+y} * \frac{x-y}{x-y} = \frac{(a+b)*(x-y)}{x^2+y^2} = \frac{(ax+by)+(bx-ay)}{x^2+y^2} = \frac{ax+by}{x^2+y^2} + \frac{bx-ay}{x^2+y^2}$$

ALGORITHM:

Step 1: start

Step 2: Read Two complex numbers c_1, c_2

Step 3: $c_3=c_1+c_2$

Step 4: print c_3

Step 5: $c_3=c_1-c_2$

Step 6: print c_3

Step 7: $c_3=c_1*c_2$

Step 8: print c_3

Step 9: $c_3=c_1/c_2$

Step 10: print c_3

Step 11: print c

Step 12: stop

PROGRAM:

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
#include<conio.h>
```

```
struct complex
```

```
{
```

```
float real,img;
```

```
};
```

```
/*code for reading complex number*/
```

```
struct complex read_complex()
```

```
{
```



```
struct complex c;  
    printf("enter real part of complex number");  
    scanf("%f",&c.real);  
    printf("enter Imaginary part of complex number");  
    scanf("%f",&c.img);  
    return c;  
}  
  
/*code for adding complex numbers*/  
struct complex add_complex(struct complex c1,struct complex c2)  
{  
    struct complex c3;  
    c3.real=c1.real+c2.real;  
    c3.img=c1.img+c2.img;  
    return c3;  
}  
  
/*code for subtraction of complex numbers*/  
struct complex sub_complex(struct complex c1,struct complex c2)  
{  
    struct complex c3;  
    c3.real=c1.real-c2.real;  
    c3.img=c1.img-c2.img;  
    return c3;  
}  
  
/*code for multiplication of complex numbers*/  
struct complex mul_complex(struct complex c1,struct complex c2)  
{  
    struct complex c3;  
    c3.real=c1.real*c2.real-c1.img*c2.img;  
    c3.img= c1.img*c2.real+c2.img*c1.real;  
    return c3;  
}  
  
/*code for division of complex numbers*/  
struct complex div_complex(struct complex c1,struct complex c2)  
{  
    struct complex c3;  
    c3.real= (c1.real*c2.real+c1.img*c2.img)/(c2.real*c2.real+c2.img*c2.img);  
    c3.img= (c1.img*c2.real-c1.real*c2.img)/(c2.real*c2.real+c2.img*c2.img);  
    return c3;  
}  
  
/*code for display of complex number*/  
void display_complex(struct complex c)  
{  
    char sign;  
    printf("The result is:");  
    if(c.img<0)
```

```
{
    sign='-';
    c.img=-c.img;
}
else
    sign='+';
printf("%5f%ci%5f",c.real,sign,c.img);

}

int main()
{
int choice;
struct complex a,b,c;
while(1)
{
printf("\n.....\n");
printf("|Menu for operation complex numbers|\n ");
printf(".....\n");
printf("1.Addition \n ");
printf("2.Subtraction \n ");
printf("3.Multiplication \n ");
printf("4.Division \n ");
printf("5.Clear Screen \n ");
printf("6.Exit Menu \n ");
printf("Enter Your Choice: ");
scanf("%d",&choice);
switch(choice)
{
case 1:printf("You Have Selected Addition operation on complex NUmbers\n");
        printf("Enter First complex number\n");
        a=read_complex();
        printf("Enter Second complex Number\n");
        b=read_complex();
        c=add_complex(a,b);
        display_complex(c);
        break;
case 2:printf("You Have Selected Subtraction operation on complex NUmbers\n");
        printf("Enter First complex number\n");
        a=read_complex();
        printf("Enter Second complex Number\n");
        b=read_complex();
        c=sub_complex(a,b);
        display_complex(c);
        break;
case 3:printf("You Have Selected Multiplication operation on complex Numbers\n");
        printf("Enter First complex number\n");
        a=read_complex();
        printf("Enter Second complex Number\n");
        b=read_complex();
```

```
        c=mul_complex(a,b);
        display_complex(c);
        break;
case 4:printf("You Have Selected Division operation on complex Numbers\n");
        printf("Enter First complex number\n");
        a=read_complex();
        printf("Enter Second complex Number\n");
        b=read_complex();
        c=div_complex(a,b);
        display_complex(c);
        break;
case 5: clrscr();
        break;
case 6: exit(0);
        default:printf("Invalid choice");
        }
    }
}
```

Record at least 3 results

Signature of faculty with date

14)b) write a c program to display the contents of a file.

Aim: To display the contents of a file.

Program:

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
#include <process.h>
int main()
{
    FILE *fs;
    char ch;
    char *fname;

    printf("Enter the file name :");
    gets(fname);
    fs = fopen(fname,"r");
    if(fs==NULL)
    {
        puts("Source file cannot be opened.");
        getch();
    }
    else
    {
        while((ch=fgetc(fs))!=EOF)
        {
            putchar(ch);
        }
    }
    getch();
    return 0;
}
```

Input:

Enter the file name :sample.txt

Output:

this is my first program

Record at least 3 results

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Week: 15

15)a) Write a C program to copy the contents of one file to another.

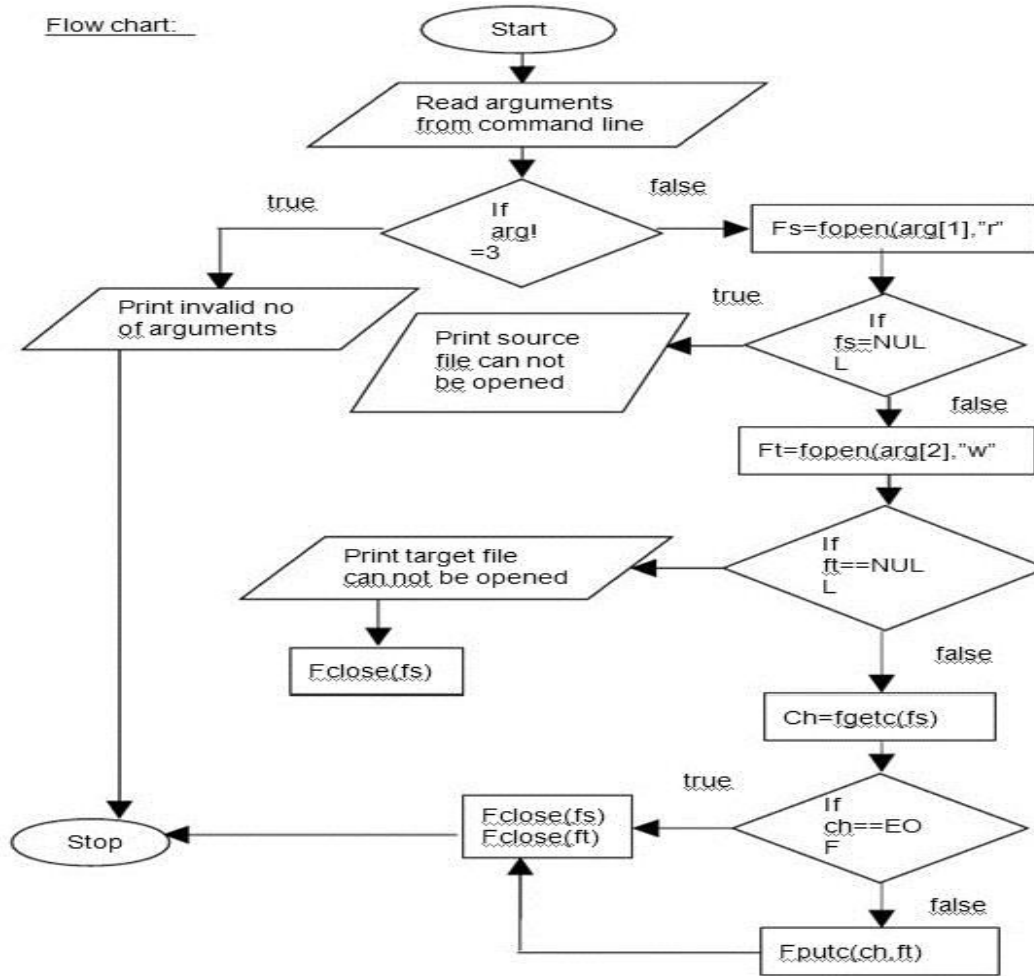
Aim:

Program which copies one file to another

Algorithm:

- Step 1: Start
- Step 2: read command line arguments
- Step 3: check if no of arguments =3 or not. If not print invalid no of arguments
- Step 4: open source file in read mode
- Step 5: if NULL pointer, then print source file can not be open
- Step 6: open destination file in write mode
- Step 7: if NULL pointer, then print destination file can not be open
- Step 8 : read a character from source file and write to destination file until EOF
- Step 9: Close source file and destination file
- Step 10: Stop

Flow chart:

**Program:**

```

#include<stdio.h>
#include<process.h>
#include<conio.h>
void main()
{
FILE *ft,*fs;
int c=0;
clrscr();
fs=fopen("a.txt","r");
ft=fopen("b.txt","w");
if(fs==NULL)
{
printf("Source file opening error\n");
exit(1);
}
else
if(ft==NULL)
{

```

```
        printf("Target file opening error\n");
        exit(1);
    }
    while(!feof(fs))
    {
        fputc(fgetc(fs),ft);
        c++;
    }
    printf("%d bytes copied from 'a.txt' to 'b.txt'",c);
    c=fcloseall();
    printf("%d files closed",c);
}
```

INPUT:

a.txt

An array is a collection of elements of similar datatypes

OUTPUT:

57 bytes copied from 'a.txt' to 'b.txt'

2 files closed

Record at least 3 results

Signature of faculty with date

15) b) Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third.

Program :

```
#include<stdio.h>
#include<conio.h>
int main()
{
    FILE *fp1,*fp2,*fp3;
    char file1[20],file2[20],file3[20],ch;
    puts("Program to merge two files... \n");
    puts("Enter first file name:");
    gets(file1);
    puts("Enter Second file name:");
    gets(file2);
    puts("Enter Destination file name:");
    gets(file3);
    fp1=fopen(file1,"r");
    fp2=fopen(file2,"r");
    fp3=fopen(file3,"w");
    if(fp1==NULL&&fp2==NULL)
        printf("Error opening file1 and file2. ... \n");
    else
    {
        if(fp3==NULL)
            printf("Error in creating destination file. .. \n");
        else
        {
            while((ch=fgetc(fp1))!=EOF)
                putc(ch,fp3);
            while((ch=fgetc(fp2))!=EOF)
                putc(ch,fp3);
        }
        printf("File Merging Sucessfull... ");
        fcloseall();
        getch();
    }
}
```

Record at least 3 results

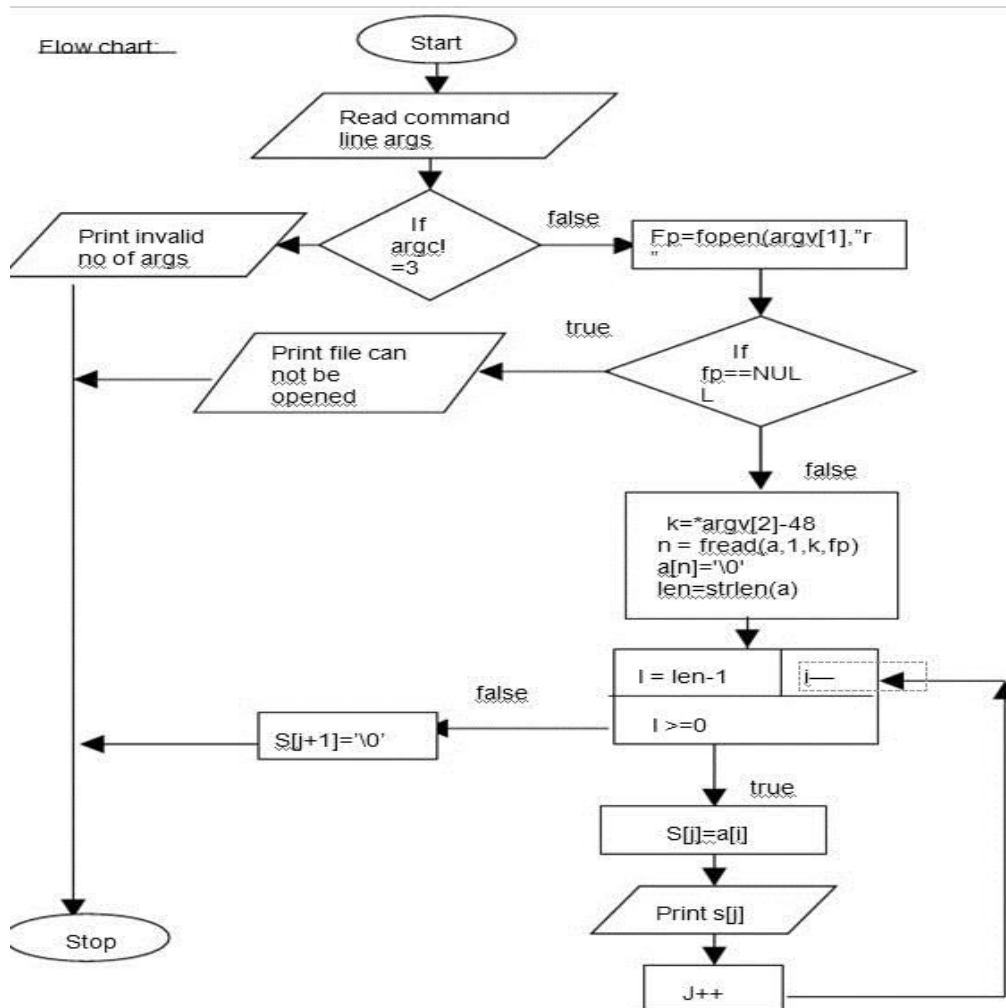
Signature of faculty with date

15) c) Write a C program to reverse the first n characters in a file.
(Note: The file name and n are specified on the command line.)

Aim: To reverse the first n characters in a file

Algorithm:

- Step 1: Start
- Step 2: read the command line arguments
- Step 3: check if arguments=3 or not
 - If not print invalid no of arguments
- Step 4: open source file in read mode
- Step 5: if NULL pointer, then print file can not be open
- Step 6: Store no of chars to reverse in k
 - $K = *argv[2] - 48$
- Step 7: read the item from file stream using fread
- Step 8: Store chars from last position to initial position in another string(temp)
- Step 9: print the temp string
- Step 10: Stop



Program:

```
#include <stdio.h>
#include <conio.h>
#include <string.h>
#include <process.h>
void main(int argc, char *argv[])
{
    char a[15];
    char s[20];
    char n;
    int k;
    int j=0;
    int i;
    int len;
    FILE *fp;
    if(argc!=3)
    {
        puts("Improper number of arguments.");
        exit(0);
    }
    fp = fopen(argv[1],"r");
    if(fp == NULL)
    {
        puts("File cannot be opened.");
        exit(0);
    }
    k=*argv[2]-48;
    n = fread(a,1,k,fp);
    a[n]='\0';
    len=strlen(a);
    for(i=len-1;i>=0;i--)
    {
        s[j]=a[i];
        printf("%c",s[j]);
        j=j+1;
    }
    s[j+1]='\0';
    getch();
}
```

Input:

source.c

 this is source

ouput.c

 this is source

Output: Command line arguments

source.c ouput.c

source.c

 this is source

ecruos si siht

Record at least 3 results

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